

Efficient Multi-Label Conformal Classification

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Abstract

In multi-label classification, the label space consists of the power set of a given set of class labels. Generating conformal classifiers for such tasks may be challenging, as the label space grows exponentially with the number of class labels, potentially resulting in a very high computational cost. Moreover, prediction sets containing label sets may be difficult to interpret; for example, the inclusion of a superset does not entail the inclusion of any of its subsets. A novel approach to multi-label conformal classification is proposed that addresses these problems by assuming an object-specific scoring function over the class labels, defining an ordering according to which labels are included in the prediction set. Results from an empirical investigation on six multi-label classification datasets, using both multiple single-target models and single multiple-target models, confirm that the theoretically guaranteed coverage is achieved and show that the choice of scoring function may have a substantial impact on predictive efficiency.

Keywords: Multi-label classification, conformal classification.

1. Introduction

In multi-label classification, the label assigned to each object is a subset, rather than a single element, of the possible class labels. Multi-label classification has gained considerable interest in the Life Science domain as a result of recent developments in machine learning including deep learning (Cerri et al., 2009; Diakou et al., 2024; Chung et al., 2023; Pliakos et al., 2021; Ren et al., 2022; Wu et al., 2021).

Conformal classifiers developed for classification tasks where a single label is assigned to each object may in principle be applied also in the multi-label setting, as their output (prediction sets) are elements of the label space. Alternatively, a prediction set may be formed by the union of non-rejected labels from a set of binary conformal classifiers, as proposed in (Wang et al., 2014). It should however be noted that the coverage guarantee of a conformal classifier developed for the single-label setting does not carry directly over to the multi-label setting where a coverage guarantee on the joint label set is required. To address this, an approach employing Bonferroni correction for the binary conformal classifiers was proposed in (Wang et al., 2015). However, this approach does not take into account any interactions among the labels, since a separate model is generated for each label, and as Bonferroni correction is known to be conservative when errors are not independent, the modified significance level may allow few labels to be rejected when handling large numbers of class labels.

An alternative approach is to rely on that the conformal prediction framework does not incur any constraints on the type of label used. The prediction set of a conformal classifier can in a multi-label setting be obtained in the same way as for the standard (single-label) classification task, i.e., by straightforwardly enumerating the elements of the (finite) label space and rejecting those for which the p-value is less than a specified significance level, as first investigated in (Papadopoulos, 2014). However, for classification tasks with large numbers of labels, the procedure may become computationally very costly, as the number of possible (multi-)labels to test grows exponentially with the number of class labels. An approach to reduce this cost was proposed in (Maltoudoglou et al., 2022), by which label sets for which the p-value in advance can be determined to not reach above the requested significance level are excluded. A more fundamental concern is that prediction sets consisting of label sets may not be the most natural and useful output. For example, without imposing any constraints on the nonconformity function, a prediction set may very well contain supersets of sets that are not included. In other words, a non-rejected label set, e.g., a set of diagnosis codes $\{D_1, D_2, D_3\}$ that is included in a prediction set may subsume a set that has been excluded, e.g., $\{D_1, D_2\}$, which may be quite counterintuitive for the practitioner, e.g., if it may be rejected that a patient suffers from $\{D_1, D_2\}$, then one would expect any superset of this to be rejected too. An arguably more natural output would be a subset of the class labels with a guarantee on subsuming the correct label set.

In this work, we propose a procedure to efficiently generate such prediction sets while maintaining a coverage guarantee. Moreover, the approach is agnostic to whether the classification task is converted to multiple binary tasks or a multi-target model is employed, allowing for interactions among the class labels to be exploited.

The main contributions of the paper are:

- An efficient procedure for multi-label conformal classification
- An empirical evaluation of the proposed approach across datasets, model types, confidence levels and nonconformity functions.

In the next section, we describe standard conformal classifiers and their extension to multi-label settings, including the novel approach. In Section 3, we describe experimental setup and results. Finally, we summarize the main findings and outline some directions for future work in Section 4.

2. Conformal classification

2.1. Conformal classifiers for single-label prediction

Given a multiset of examples $\{z_1, \dots, z_{q+1}\}$, where each $z_i \in \mathcal{Z}$, for some example space $\mathcal{Z}$, we may obtain a multiset of nonconformity scores $\{\alpha_1 = A(z_1), \dots, \alpha_{q+1} = A(z_{q+1})\}$ using some nonconformity function $A : \mathcal{Z} \rightarrow \mathbb{R}$.

From the multiset of nonconformity scores $\{\alpha_1, \dots, \alpha_{q+1}\}$ and a value τ that is sampled uniformly from the unit interval, a (smoothed) p-value for the $q + 1$ th nonconformity score is computed by:

$$p_{q+1} = \frac{|i : \alpha_i > \alpha_{q+1}| + \tau |i : \alpha_i = \alpha_{q+1}|}{q + 1} \quad (1)$$

where $i \in \{1, \dots, q+1\}$ (Vovk et al., 2022).

For prediction tasks, each example z_i consists of a pair $(\mathbf{x}_i, y_i)$, where $\mathbf{x}_i \in \mathcal{X}$, for some object space $\mathcal{X}$, and $y_i \in \mathcal{Y}$ for some label space $\mathcal{Y}$. For (standard) classification tasks, the label space is finite, i.e., $\mathcal{Y} = \{c_1, \dots, c_k\}$. We will in this work focus on the inductive conformal prediction setting, in which a given training set is randomly split into a proper training set Z_{train} and a calibration set Z_{cal} , and where the nonconformity function A is defined using the proper training set.

Given a calibration (multi-)set $Z_{cal} = \{z_1, \dots, z_q\}$, a nonconformity function A , a significance level ϵ , and a test object $\mathbf{x}_{q+1}$, an (inductive) *conformal classifier* outputs a prediction set:

$$C = \{c \in \mathcal{Y} : z_{q+1} = (\mathbf{x}_{q+1}, c) \text{ \& } p_{q+1} \geq \epsilon\} \quad (2)$$

Let $z_{q+1} = (\mathbf{x}_{q+1}, y_{q+1})$ where y_{q+1} is the true class label for the object $\mathbf{x}_{q+1}$. Under the assumption that the examples $\{z_1, \dots, z_{q+1}\}$ are exchangeable, it follows that p_{q+1} is distributed uniformly on $[0, 1]$ (Vovk, 2021). This gives the following (exact) coverage guarantee:

$$\mathbb{P}(y_{q+1} \in C) = 1 - \epsilon \quad (3)$$

If a coverage guarantee is required for each label, we may form a label-conditional Mondrian conformal classifier (Löffström et al., 2015), by partitioning the calibration set $Z_{cal} = \{z_1, \dots, z_q\}$ into subsets $Z_1, \dots, Z_k$, where each subset Z_m is defined by:

$$Z_m = \{(\mathbf{x}_i, y_i) : (\mathbf{x}_i, y_i) \in Z_{cal} \text{ \& } y_i = c_m\} \quad (4)$$

The coverage guarantee (Eq. 3) still holds for each category (class label) under the assumption of the exchangeability for $\{z_1, \dots, z_{q+1}\}$, as exchangeability is implied for all subsets, including the subset that contains all examples that have been assigned to a specific category.

2.2. Conformal classifiers for multi-label prediction

Since the power set of any finite set of class labels is finite, we may form a conformal classifier using the label space $\mathcal{Y} = \mathcal{P}(\{c_1, \dots, c_k\})$ and generate prediction sets in the same way as for conformal classifiers for single-label tasks, i.e., by enumerating the elements of the label space and including those for which the corresponding p-value is not less than the specified significance level. However, this approach may be prohibitively expensive in practice, as the number of labels to test (2^k) grows exponentially with the number of class labels. Another drawback of this approach is that the elements of the label space are handled independently, and if no constraints are imposed on the nonconformity scores, we may very well reject the inclusion of a subset in the prediction set, while including a superset.

To address these problems, we propose to use an (object-specific) scoring function for the class labels $\pi_{\mathbf{x}} : \{c_1, \dots, c_k\} \rightarrow \mathbb{R}$ in a nonconformity score function on the following form:

$$A(z = (\mathbf{x}, y)) = 1 - \min_{\ell \in y} \pi_{\mathbf{x}}(\ell) \quad (5)$$

Let $\alpha_{(1)}, \dots, \alpha_{(q)}$ be the nonconformity scores for the calibration set, sorted in ascending order, and let $\tau_\epsilon = \alpha_{(\lceil (q+1)(1-\epsilon) \rceil)}$, for some significance level ϵ . Assuming a test example $z_{q+1} = (\mathbf{x}_{q+1}, y_{q+1})$ with nonconformity score α_{q+1} , then under the assumption of exchangeability:

$$\mathbb{P}(\alpha_{q+1} \leq \tau_\epsilon) \geq 1 - \epsilon \quad (6)$$

This can be rewritten as:

$$\mathbb{P}(\min_{\ell \in y_{q+1}} \pi_{\mathbf{x}_{q+1}}(\ell) \geq 1 - \tau_\epsilon) \geq 1 - \epsilon \quad (7)$$

The prediction set for the test object $\mathbf{x}_{q+1}$ then becomes:

$$\Gamma^\epsilon(\mathbf{x}_{q+1}) = \{\ell : \ell \in \{c_1, \dots, c_k\} \ \& \ \pi_{\mathbf{x}}(\ell) \geq 1 - \tau_\epsilon\} \quad (8)$$

In order to generate a prediction set for a test object, a single pass over the class labels is hence required only, in which each label is tested for inclusion, by comparing the score computed using the scoring function to the threshold derived from the nonconformity scores of the calibration set. Assuming the cost of computing a score to be constant, the computational complexity of generating a prediction set is hence linear in the number of class labels, i.e., $\mathcal{O}(k)$.

The proposed nonconformity score ensures consistency with respect to subset inclusion. In particular, let $y' \subseteq y$ be two nonempty candidate label sets. Since the minimum is taken over a smaller set for y' , it follows that:

$$\min_{\ell \in y} \pi_{\mathbf{x}}(\ell) \leq \min_{\ell \in y'} \pi_{\mathbf{x}}(\ell), \quad (9)$$

and hence:

$$A(\mathbf{x}, y') \leq A(\mathbf{x}, y) \quad (10)$$

Consequently, if y is not rejected, i.e., $A(\mathbf{x}, y) \leq \tau_\epsilon$, then $A(\mathbf{x}, y') \leq \tau_\epsilon$, so y' is not rejected either. Thus, unlike an unconstrained conformal classifier over the power-set label space, the proposed approach cannot reject a subset while retaining one of its supersets.

The proposed nonconformity function bears some resemblance to the maxT statistic used in resampling-based multiple testing ([Westfall and Young, 1993](#)). In particular, defining the label-wise nonconformity score $T_\ell(\mathbf{x}) = 1 - \pi_{\mathbf{x}}(\ell)$, we have

$$A(\mathbf{x}, y) = \max_{\ell \in y} T_\ell(\mathbf{x}), \quad (11)$$

where each example is represented by the most extreme, or least conforming, score among its true labels. Whereas maxT procedures control the probability of at least one false rejection of

the multiple hypotheses by finding a common threshold from the resampling distribution of the maximum test statistic, the proposed approach derives the threshold from the empirical distribution of the nonconformity scores for the calibration examples under the assumption of exchangeability.

The choice of scoring function does not affect the validity of the approach, but may have an effect on the predictive efficiency, i.e., the size of the resulting prediction set. In the experimental investigation, we will explore the following three scoring functions, given a set of predicted probabilities $\{p_1, \dots, p_k\}$ for a test object $\mathbf{x}$:

$$\pi_{\mathbf{x}}(\ell) = p_{\ell} \quad (\text{last})$$

$$\pi_{\mathbf{x}}(\ell) = \frac{\sum_{p_i \in \widehat{\ell}} p_i}{|\widehat{\ell}|} \quad (\text{mean})$$

$$\pi_{\mathbf{x}}(\ell) = \sum_{p_i \in \widehat{\ell}} p_i \quad (\text{sum})$$

where $\widehat{\ell} = \{p_i \in \{p_1, \dots, p_k\} : p_i \geq p_{\ell}\}$.

3. Experimental Investigation

In this section, we evaluate the proposed approach for multi-label conformal classification. We first describe the experimental setup and then present and discuss the findings.

3.1. Experimental Setup

3.1.1. DATASETS

The following datasets were selected for the investigation: Two of the ToxCast datasets in MoleculeNet ([Wu et al., 2017](#)), consisting of 7 933 compounds with 8 and 15 endpoints, respectively, representing agonists and antagonists, SIDER, also from MoleculeNet, consisting of 1 427 compounds and 27 indications, the kinase dataset HTSFP10 ([de la Vega de León et al., 2018](#)), consisting of 56 892 compounds and 10 endpoints¹, the yeast_ml8 dataset ([OpenML, a](#)), consisting of 2 417 examples and 14 endpoints, and finally the Reuters news articles dataset ([OpenML, b](#)), consisting of 2000 articles and 7 categories.

3.1.2. METHODS

RDKit (v2023.09.5) ([Landrum](#)) was used to calculate features from SMILES strings for the first four datasets. See supplementary information in ([Svensson et al., 2017](#)) for a list of used descriptors.²

A RandomForestClassifier (with default settings) from Scikit Learn (v 1.7.2) ([Pedregosa et al., 2011](#)) was used to build predictive models in two different ways; by generating a binary model for each class label and by generating a joint (multi-target) model for all labels.

1. Missing labels for 15 instances were changed to inactive (majority class) labels.

2. The Ipc descriptor was not included.

Table 1: Average coverage using multiple single-target models.

conf.	last	mean	sum
0.70	0.7922	0.7969	0.7994
0.75	0.8247	0.8257	0.8195
0.80	0.8604	0.8660	0.8544
0.85	0.8966	0.9012	0.8888
0.90	0.9381	0.9373	0.9221
0.95	0.9685	0.9708	0.9619

The predicted probabilities $\mathbf{p}_x = \{p_1, \dots, p_k\}$ for an object x were used to generate scores $\pi_x(\ell_1), \dots, \pi_x(\ell_k)$, with each of the three scoring functions in Section 2.2, which in turn is used to compute non-conformity scores (Eq. 5). When employing a multi-target model, all the predicted probabilities for a test object are obtained at once, while when using a binary model for each label c_i , each probability p_i corresponds to the predicted probability of the positive label.

3.1.3. TEST PROTOCOL

Each dataset was randomly split into a training and a test set (50%/50%). The training set was subsequently split into a proper training set and a calibration set (75%/25%). The employed confidence levels were $\{0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$

For each conformal predictor, the coverage rate on the test set was recorded together with the relative efficiency, measured as the average number of labels in the prediction sets divided by the total number of class labels.

3.2. Experimental Results

In Table 1, the average coverage across the datasets, confidence levels and scoring functions is shown for the case when a (binary) model was generated for each target. It can be seen that the observed error rates did not exceed the significance (1-confidence) levels on average, as expected from the theoretical guarantee (Eq. 6), but also that the observed coverage was generally quite conservative, in particular for the lower levels of confidence.

In Table 2, the average coverage is reported for the case when a single multi-target model was employed. It can be seen again that the coverage tend to be on the conservative side. The scoring function **sum**, which tended to be slightly less conservative than the two other functions when using multiple single-target models, in contrast is the most conservative for the multi-target models.

The average relative efficiency when using multiple single-target models is shown in Table 3. In contrast to the observations on coverage, the results on efficiency for the three scoring functions were clearly separated; **last** resulted in smaller prediction sets than **mean**, which in turn generated smaller prediction sets than **sum**. For the highest level of confidence, **last** rejected almost 30% of the labels on average, while the worst-performing approach (**sum**) rejected just above 5% of the labels on average.

Table 2: Average coverage using a single multi-target model.

conf.	last	mean	sum
0.70	0.7844	0.7870	0.8228
0.75	0.8158	0.8147	0.8496
0.80	0.8511	0.8473	0.8749
0.85	0.8852	0.8843	0.8959
0.90	0.9219	0.9284	0.9324
0.95	0.9617	0.9652	0.9682

Table 3: Average relative efficiency using multiple single-target models.

conf.	last	mean	sum
0.70	0.4148	0.4442	0.4817
0.75	0.4490	0.4795	0.5200
0.80	0.4899	0.5391	0.6150
0.85	0.5458	0.5969	0.7033
0.90	0.6245	0.6764	0.8505
0.95	0.7131	0.7941	0.9430

The relative performance of the three scoring functions remained the same when using a single multi-target model, as can be seen in Table 4. Comparing the results across model types, one may observe a minor gain from using the multi-target model instead of multiple single-target model, for the two best performing scoring functions (**last** and **mean**), across all considered levels of confidence.

In Fig. 1 and 2, we take a closer look at the results for 80% confidence, which is typical level for many machine learning models that operate in the Life Sciences domain.

In Fig. 1, box plots for the ratio between the average number of labels in the prediction sets and the average size of the correct label sets are shown, for the three scoring functions and two underlying models. Also for this metric, **last** was the best performing scoring function on average, while the difference between the underlying models are visible only for the less performant functions. Although the overall difference between the two types of

Table 4: Average relative efficiency using a single multi-target model.

conf.	last	mean	sum
0.70	0.4056	0.4433	0.4809
0.75	0.4382	0.4865	0.5249
0.80	0.4846	0.5305	0.5961
0.85	0.5341	0.5932	0.6567
0.90	0.5937	0.6613	0.8502
0.95	0.7102	0.7973	0.9465

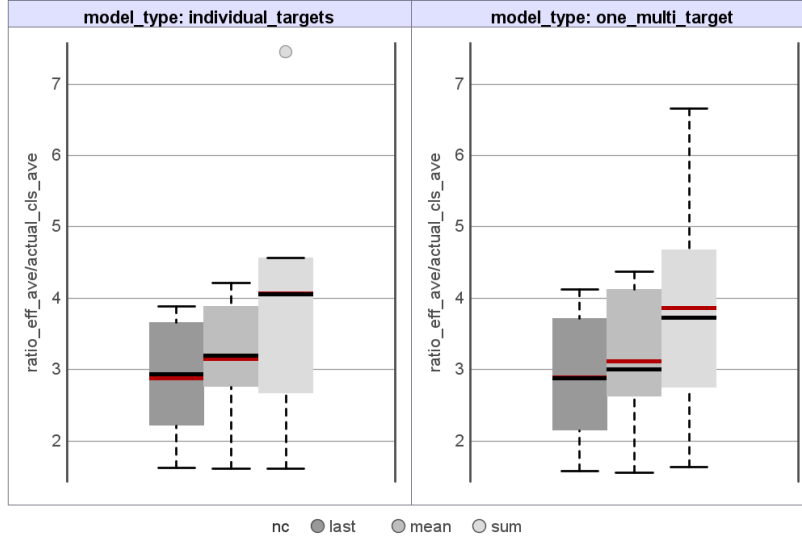


Figure 1: Ratio between averages of predicted and correct number of labels across datasets at the 80% confidence level (black line = median value, red line = mean value).

model is rather small on average, a closer look shows that the scoring function **sum** resulted in a larger deviation of prediction sizes compared to the other two (**last** and **mean**), in particular for the multi-target models.

A similar observation was made also for coverage, as shown in Fig. 2, where a larger span can be noticed for the scoring function **sum**. However, the number of observations was too small to allow for drawing conclusions that the observed differences are statistically significant (using Anova + Tukey’s HSD at $p < 0.05$).

4. Concluding Remarks

We have presented a computationally efficient procedure for multi-label conformal classification, which instead of evaluating subsets of the set of class labels, performs a single pass to test each class label for inclusion in the prediction set, using a scoring function. The coverage guarantee follows from employing an appropriately defined general form of non-conformity function, which allows any scoring function to be used.

Results from an empirical investigation have been presented, in which the resulting conformal classifiers were shown to indeed meet the requested coverage rates. The proposed approach is applicable when using a single multi-target model as well as multiple single-target models. Although the former may in contrast to the latter exploit interactions among the class labels, both types of underlying model were observed to result in similar performance for the considered datasets. Three scoring functions were evaluated and the simplest of them, which uses the predicted probability as a score, was shown to yield the smallest prediction sets.

There are several directions for future research. An immediate extension of the study is to include more datasets to increase power of the statistical test when comparing different

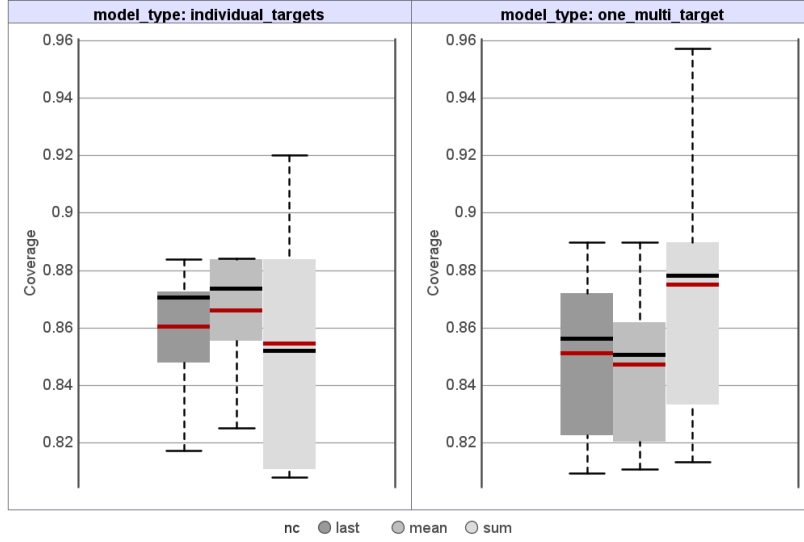


Figure 2: Coverage across datasets at the 80% confidence level (black line = median value, red line = mean value).

approaches. A comparison to approaches that employ Bonferroni correction or evaluate sets of labels could put light on differences in computational cost and predictive efficiency in practice, although a prediction set that consists of a subset of the class labels, as generated by the proposed and Bonferroni-corrected approaches, is not directly comparable to output in the form of a set of subsets of class labels. In the current study, three rather straightforward scoring functions were investigated. An important direction for future work is to propose and evaluate more sophisticated scoring functions.

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